

scRNA-seq Batch Effects & Bulk RNA-seq Integration

A Structured Research Summary

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Three reports → one reference document: batch effect biology, bulk–single-cell methods, and bulk-informed calibration.

EXECUTIVE SUMMARY

This document consolidates three research reports into a single structured reference. Key findings:

- **Calibration crisis:** 7 of 8 widely used batch correction methods introduce artifacts even when no batch effect exists. Only Harmony passes null-condition calibration tests consistently.
- **Pseudobulk is the standard:** Pseudobulk + DESeq2/edgeR is mathematically equivalent to GLMMs (proved 2024) and avoids the 549-fold false-discovery inflation of naïve single-cell DE.
- **557 discordant genes:** Matched bulk–scRNA-seq comparisons reveal hundreds of genes with systematic technology-specific bias. Removing these improves cross-platform analysis.
- **Pilot pairing strategy:** Generating matched bulk for just 6 donors provides disproportionate calibration returns—outperforming much larger external references.
- **Critical gap:** No widely adopted method uses bulk-derived priors (GTEx, TCGA, ENCODE) for scRNA-seq batch correction. This is a clear research opportunity.

PART I: BATCH EFFECTS IN scRNA-SEQ

1.1 Sources of Systematic Bias

Every step from tissue to sequencer introduces systematic variation:

- **Platform:** Smart-seq2 detects >25K genes/sample (full-length); droplet methods capture only 3' ends with fewer genes. Certain cell types vanish entirely from some platforms.
- **Chemistry version:** 10x v4 (GEM-X) captures 98% more genes and 249% more transcripts per cell vs. v3.1.
- **Sequencing depth:** PC1 correlates with detection rate ($r = 0.54\text{--}0.93$ across 15 studies) —the dominant variation axis is often technical, not biological.
- **Dissociation:** ~512 stress-response genes (FOS, JUN, EGR1, HSPA1A) activated at 37°C. Creates fictitious cell clusters (e.g., artificial “activated microglia” in hippocampal tissue).
- **Ambient RNA:** 2–50% of UMI counts per cell (typically ~10%). Batch-specific contamination creates false subclusters and spurious DE genes.
- **Reagent lots:** One lot change in a clinical trial caused 162 incorrect patient classifications, 28 receiving wrong chemotherapy.
- **Biological confounders:** Donor variation, age, sex, circadian rhythms confounded with processing order.

1.2 Why Single-Cell Amplifies Batch Effects

- **Sparsity:** ~80% zero entries. Dropout rates are batch-dependent, creating bias in the zero-inflation structure with no bulk analog.
- **Low sampling:** 1–5K genes, 2–30K UMIs per cell vs. millions in bulk. Each cell is an unrepeatable single measurement.
- **Composition differences:** Batches contain non-identical cell-type proportions due to protocol biases. Algorithms cannot distinguish compositional biology from technical artifact.

1.3 Downstream Damage

- **Clustering:** Cells segregate by lab/technology, not cell type. scIB benchmark: 85 batches, 1.2M cells confirm within-batch distances < between-batch distances.
- **Differential expression:** First Alzheimer’s snRNA-seq study: 1,031 DEGs reported → ~16 robust DEGs after proper pseudobulk reanalysis (549-fold reduction).
- **Trajectory inference:** Dissociation artifacts create spurious branches; sequencing depth shifts cells along pseudotime.
- **Gene regulatory networks:** Dropout creates false co-expression. Existing GRN methods already perform at random-predictor levels.

1.4 Correction Methods at a Glance

Method	Space	Scale	Calibration	Key Tradeoff
Harmony	PCA	Excellent	Good	Only well-calibrated method; linear—

				degrades on complex cross-species tasks
Seurat CCA/RPCA	CCA → expr.	Moderate	Moderate	Overcorrects when cells lack cross-dataset counterparts
scVI	Latent (counts)	Good (GPU)	Overcorrects	Alters data even under null conditions; sensitive to hyperparameters
scANVI	Latent (counts)	Good (GPU)	Good	Top overall in scIB; requires partial labels
Scanorama	SVD → expr.	Moderate	Good	Avoids forcing non-overlapping populations together
MNN/fastMNN	Expr./PCA	Poor	Poor	Sensitive to merge order; poorly calibrated in 2025 evaluation
ComBat	Expression	Excellent	Moderate	Assumes same composition across batches; can't handle sparsity

Best Practice

Use **embedding-based correction** (Harmony) for clustering/visualization. Run DE on **original uncorrected counts** with batch as covariate—never on corrected expression matrices.

1.5 Overcorrection vs. Undercorrection

- **Overcorrection:** Merges distinct cell types, erases disease states, collapses gradients. Signature: ribosomal proteins dominate cluster markers.
- **Undercorrection:** Cells cluster by sample, “novel” populations are batch artifacts, inflated DE results.
- **Mathematically intractable:** When batches contain non-overlapping cell types, no algorithm can separate batch from biology (formal proof exists).
- **Metrics are unreliable:** Silhouette-based measures (used in 100+ papers) are fundamentally flawed for scRNA-seq. BRAS and RBET (2025) are proposed replacements.

1.6 2024–2025 Developments

- **Calibration test (2025):** Random pseudo-batch splitting shows 7/8 methods create artifacts under null conditions. Only Harmony passes.
- **Foundation models:** scGPT and Geneformer perform worse than Harmony/scVI for batch integration without fine-tuning. Established methods remain competitive.
- **Atlas-scale paradigm:** Build core reference with scANVI/scVI, then map new data via scArches (fine-tunes query parameters) or Symphony (72K cells mapped to 1.39M-cell reference in 11 seconds).

PART II: HOW BULK RNA-SEQ SHAPED SINGLE-CELL ANALYSIS

2.1 Deconvolution

Estimating cell-type fractions in bulk tissue using single-cell references:

- **53+ methods** catalogued (2024 review). Span nu-SVR, weighted NNLS, deep learning, and Bayesian frameworks.
- **Reference quality > algorithm choice.** Protocol-matched in-study references (as few as 6 donors) outperform much larger external references.
- **DeMixSC (2025):** Calibrates with small matched bulk/sc benchmark. Partitions genes into concordant (G1) and discordant (G2), rescales G2. MAE = 0.03, Spearman = 0.75.
- **557 discordant genes** identified in matched BAL data (Hu et al. 2026)—removing these improves deconvolution at fine cell-type resolution.

2.2 Pseudobulk DE

The most consequential bulk → single-cell insight:

- **Problem:** Treating cells as independent replicates inflates false positives catastrophically (pseudoreplication).
- **Solution:** Sum raw counts per cell type per sample → apply DESeq2/edgeR/limma-voom to pseudobulk profiles.
- **Proved in 2024:** Pseudobulk with proper offsets is mathematically equivalent to GLMMs—same point estimates, same standard errors, much faster.
- **Limitation:** Degrades with <3 replicates/condition. Cannot detect rare cell-state transitions within a type.

2.3 Normalization Adaptations

- **Bulk methods break:** DESeq2 median-of-ratios relied on as few as 115 genes in scRNA-seq (unstable). TMM interacts pathologically with sparsity.
- **scran (2016):** Pools similar cells, computes pool-level size factors (zeros diluted by summation), deconvolves back to individual cells.
- **sctransform (2019):** Regularized negative binomial regression per gene. Pools information across genes with similar mean expression for stable dispersion estimates.
- **Best practice:** scran for count-based analyses; sctransform Pearson residuals for dimensionality reduction/visualization.

2.4 Bulk Atlases for Annotation

- **GTEx** (53 tissues, ~1K individuals), **TCGA** (33 cancer types, ~11K patients), **ENCODE** provide the biological knowledge base for scRNA-seq interpretation.
- **SingleR:** Annotates single-cell clusters by correlation against bulk reference profiles.
- **Transfer learning:** Models pre-trained on GTEx successfully adapted for single-cell cell-type annotation (Park et al. 2021).

- **Foundation models** (scGPT, 33M+ cells) implicitly benefit from gene function annotations established through decades of bulk genomics.

2.5 Gene Signature Scoring

- **Problem:** ssGSEA and GSVA are biased by number of detected genes per cell—cancer cells get inflated scores regardless of true pathway activity.
- **Solution:** Use ranking-based methods designed for scRNA-seq: AUCell, JASMINE, UCell.
- **irGSEA (2024):** Unified framework integrating six rank-based scoring methods for systematic comparison.

2.6 Imputation with Bulk Priors

- **SCRABBLE:** Convex optimization constrained so aggregated imputed data matches bulk means. At 72% dropout, outperforms MAGIC, scImpute, VIPER, SAVER.
- **scDTL (2024):** Denoising autoencoder pre-trained on bulk RNA-seq, then domain adaptation via MMD loss to align with single-cell embeddings.
- **Caution:** Svensson (2020) argued droplet data is not zero-inflated—many “dropouts” are expected sampling zeros. Imputation risks creating false gene-gene correlations.

2.7 Long-Read Isoform Analysis

- **Gap:** Standard 10x captures only 3' ends—alternative splicing is invisible.
- **Bulk infrastructure:** GENCODE, RefSeq, SQANTI annotations built from bulk long-read RNA-seq.
- **FLAMES/FLAMESv2:** Single-cell isoform analysis via 10x + Nanopore. Detects 3–14× more novel junctions than short reads.
- **Longcell (2025):** Statistical framework for isoform quantification from spatially barcoded Nanopore reads.

PART III: BULK AS CALIBRATION LENS FOR sc BATCH EFFECTS

3.1 Matched Bulk Exposes Hidden Artifacts

- **Hu et al. (2026)**: 30 BAL donors, >240K cells. 557 reproducibly discordant genes ($|\log_2FC| > 1$, $FDR < 0.05$) between real bulk and matched pseudobulk.
- **PC1 = technology, not biology**. Sparse PCA separates samples by modality (bulk vs. pseudobulk); biological heterogeneity only on PC2.
- **Pseudobulk-enriched genes**: Mitochondrial function, cytoplasmic translation (intracellular processes).
- **Bulk-enriched genes**: Cell-cell communication, ECM interactions, tissue-level dynamics lost during dissociation.
- **Squair et al. (2021)**: 18 matched datasets confirm single-cell DE methods produce systematic false positives for highly expressed genes.

3.2 Calibration Methods

DeMixSC (Genome Research, 2025)

- Requires small matched bulk/sc benchmark dataset.
- Partitions genes into G1 (concordant) and G2 (discordant); rescales G2 before deconvolution.
- MAE = 0.03, Spearman $r = 0.75$ on 24 matched retinal samples.

CIBERSORTx: S-mode (adjusts sc signatures $\rightarrow$ bulk characteristics) and B-mode (adjusts bulk $\rightarrow$ full-length protocol signatures).

ReDeconv (2025): Introduces CLTS normalization preserving transcriptome size differences between cell types. Corrects three bias types: scaling (Type-I), gene length (Type-II), variance (Type-III).

Bisque: Learns gene-specific transformations between bulk and single-cell. $R = 0.923$ on matched adipose tissue data.

3.3 Transfer Learning: Bulk $\rightarrow$ Single-Cell

- **SAVER-X (2019)**: Deep autoencoder pre-trained on large scRNA-seq + Bayesian shrinkage. Cross-species transfer works (mouse $\rightarrow$ human).
- **scDTL (2024)**: Denoising autoencoder trained on bulk RNA-seq first. Domain adaptation via MMD loss. Outperforms all baselines at 40% dropout across 8 datasets.
- **SCRABBLE**: Convex optimization with bulk constraint (aggregated imputed data must match bulk means).
- **Bubble, scINRB, SIMPLEs-B**: Various autoencoder/NMF approaches constrained by bulk expression data.

3.4 Classic Bulk Methods Adapted for sc

- **ComBat-seq:** Negative binomial extension. TPR = 0.89, but achieves little cross-technology mixing—underperforms Harmony, Seurat 3.
- **scMerge (RUV adaptation):** Identifies 1,076 human / 826 mouse stably expressed genes (scSEGs) as negative controls. Uses fastRUVIII with pseudo-replicates.
- **RUV-III-NB:** RUV extended to negative binomial for scRNA-seq, returning adjusted counts for downstream DE/pseudotime.
- **ZINB-WaVE:** Zero-inflated NB generalization of RUV. Posterior weights unlock bulk tools (edgeR, DESeq2) for single-cell applications.

3.5 Bulk Resolves Biology vs. Artifact

Bulk RNA-seq avoids dissociation, captures full-length transcripts, and has well-characterized noise:

- **Dissociation proof:** smFISH shows Fos in 27/75 dissociated muscle stem cells vs. 0/80 in intact muscle (Van den Brink 2017).
- **512 stress genes** defined by O’Flanagan et al. (2019) across 155K cells.
- **Denisenko et al. (2020):** 15 bulk profiles + 77K sc + 98K sn from mouse kidney establish that stress induction stems from dissociation protocol itself.
- **Platform bias:** 10x dropout ratios 1.3–1.4× higher than Smart-seq2; Smart-seq2 pseudobulk more closely resembles real bulk.

3.6 Spike-ins: Limited Utility

- **BASiCS:** Uses ERCC spike-in concentrations to estimate capture efficiencies, decomposing technical vs. biological variability.
- **Limitations (Tung et al. 2017):** ERCC counts differ significantly between batches ($P < 0.001$) and across individuals ($P = 0.08$). Biased estimates expected.
- **Impractical for droplet platforms:** Variable droplet volumes, competition for reads, <46/92 ERCCs detected per cell.

3.7 Experimental Design: The Pilot Pairing Strategy

Recommended Protocol

Generate scRNA-seq for a minimal subset (~6 donors) of bulk samples. Construct in-study reference. Identify discordant genes. Apply corrections to full cohort. Achieves substantial calibration at modest cost.

- **Tung et al. design:** Each sc batch includes matching bulk from same reagents → direct quantification of technical variation.
- **BUSseq reference panel:** Common reference in every batch guarantees valid separation of batch from biology—bulk can serve as that reference.
- **Chain-type designs:** Adjacent batches share overlapping cell types. Mathematically valid correction foundation.

3.8 Emerging Frontier

- **Foundation models (scGPT, scPRINT):** Zero-shot batch correction, but none incorporate bulk pre-training—a notable gap.
- **ComBat-ref (2024):** Selects lowest-dispersion batch as anchor; naturally supports using a bulk batch as reference.
- **RBET (2025):** Uses reference gene expression patterns (potentially bulk-derived) to evaluate correction quality with overcorrection sensitivity.
- **Li et al. (Nature Computational Science, Feb 2026):** Calls for informed modeling of technical vs. biological variation—implicitly advocating bulk-informed approaches.

SYNTHESIS AND RECOMMENDATIONS

Three Overarching Findings

1. **Most correction methods are poorly calibrated.** 7/8 create artifacts under null conditions. Only Harmony is safe as a default. Evaluation metrics (silhouette-based) are themselves unreliable.
2. **Bulk and single-cell are symbiotic, not successive.** Bulk provides statistical rigor (pseudobulk DE), knowledge bases (pathways, isoforms, atlases), and validation ground truth. Single-cell provides cellular resolution.
3. **The critical gap is a bulk-informed batch correction framework.** No integrated tool uses bulk-derived gene-level variance, expression distributions, or cross-platform bias models as priors for scRNA-seq integration.

Immediate Actions

4. Use **Harmony** for routine batch correction; **scANVI** for atlas-scale integration with partial labels.
5. Perform DE via **pseudobulk + DESeq2/edgeR**—never on corrected expression matrices.
6. Adopt the **pilot pairing strategy**: matched bulk for ~6 donors of your scRNA-seq samples.
7. Replace ssGSEA/GSVA with **AUCell or UCell** for gene signature scoring.
8. Remove/downweight the **~557 known discordant genes** in cross-platform analyses.
9. Prioritize **experimental design** (randomization, multiplexing, chain-type) over computational correction.

Research Opportunities

- Develop frameworks using bulk-derived gene-level variance estimates as priors for scRNA-seq batch correction.
- Incorporate bulk pre-training into single-cell foundation models.
- Build standardized discordant gene databases across tissues and platforms.
- Adopt BRAS and RBET metrics to replace silhouette-based evaluation.